

Brian Borbath

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SUMMARY

Electrical Engineering freshman at the University of Florida passionate about semiconductor devices and hardware engineering with hands-on experience in RF, embedded systems, and Python programming.

EDUCATION

B.S. in Electrical Engineering

Expected May 2030

University of Florida • Gainesville, FL

Relevant Coursework: Calculus III, C++ for Engineers, Applied Chemistry

TECHNICAL PROJECTS

Raspberry Pi Thermal Performance Research

- Designed and conducted an experimental investigation evaluating whether 3D-printed cases could improve the thermal performance of a Raspberry Pi running AI workloads.
- Developed Python-based workflows to collect performance data and automate statistical analysis using ANOVA.

VHF Antenna Design/Testing

- Designed and tested VHF antennas for amateur radio operation using wavelength-based element calculations and SWR measurements.

Home Server & Network Infrastructure

- Configured Linux/Windows servers, VPN routing, DNS, reverse proxies, and TLS services.

Programmable Semiconductor Curve Tracer (In Progress)

- Designing a programmable semiconductor curve tracer to analyze the I-V characteristics of diodes and BJTs.
- Integrating an ESP32 microcontroller to automate data collection and transmit measurements to a PC for Python-based characterization.

EXPERIENCE

Gator Amateur Radio Club

August 2026 – Present

Secretary • UF

- Maintain meeting records, manage the club website, assist with club communications, and coordinate events such as POTAs and group meets.

The Forge

June 2025 – August 2025

Lab Assistant • Patrick Space Force Base

- Contributed to innovation efforts by supporting laboratory projects focused on reducing U.S. Space Force operational costs using 3D printers and 3D scanners.

Civil Air Patrol, USAF Auxiliary

September 2022 – June 2026

Cadet Deputy Commander • Satellite Beach, FL

- Led and mentored 75+ cadets while supervising staff through the squadron chain-of-command.
- Planned weekly meetings, training activities, and events in coordination with adult leadership.
- Dedicated 4+ hours per week to emergency services, aerospace education, and community service.

TECHNICAL SKILLS

Programming: Python, Java

Tools/Platforms: Git, Linux, VS Code, Gradle, Raspberry Pi, Microsoft Office

Engineering: RF antenna design and testing, soldering, 3D printing, electronics prototyping